

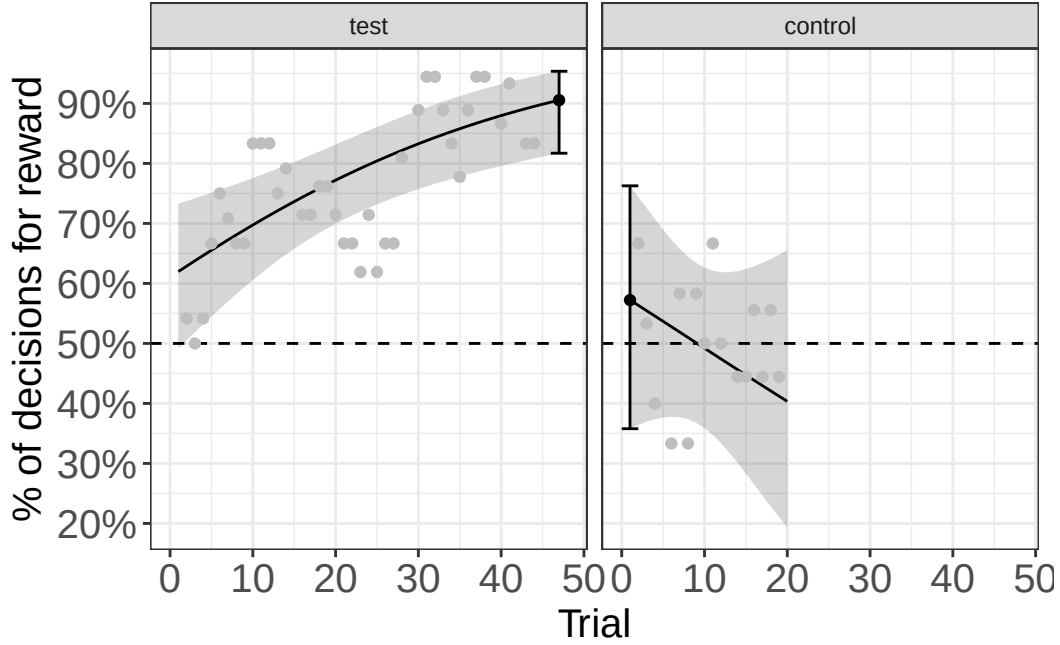
glm_analysis

Make the computations necessary for all questions:

	contrast	odds.ratio	SE	df	null	z.ratio
1	test_end / control_start	2.161893	1.3957117	Inf	1	1.1942185
2	test_end / control_start	1.302833	0.5337640	Inf	1	0.6457023
3	test_end / control_start	2.921793	1.3532234	Inf	1	2.3150201
4	test_end / control_start	0.649945	0.3150206	Inf	1	-0.8889584
5	test_end / control_start	7.156515	3.9704981	Inf	1	3.5472092
6	test_end / control_start	7.059280	3.7730719	Inf	1	3.6565048
	p.value	experiment	control_start	test_end		
1	0.2323924959	natcan1_170923	1	119		
2	0.5184721949	natcan3_080825	1	64		
3	0.0206118297	natcan3_190925	1	131		
4	0.3740254126	natcan4_180825	1	100		
5	0.0003893353	perpPara_170725	1	47		
6	0.0002556776	thin0b_150823	1	66		

Question 1: Can wasps use dorsal patterns?

- Looking at test versus control proportion correct
- Slope in test and control
- EMmeans at end of test and control
 - is this essentially equivalent to binning the last n trials and making the comparison



[1] "0.817-0.954"

		contrast	logOddsRatio	SE	df	asympt.LCL	asympt.UCL	z.ratio
1	test_end - control_start		1.968	0.555	Inf	0.881	3.055	3.547
	p.value	experiment	control_start	test_end	pFmt	confint		
1	0.000	perpPara_170725	1.000	47.000	<0.001	0.881-3.055		

During the test phase of the experiment, trial number (experience) had a significant positive effect on the expected probability (P_{corr}) of a correct decision (Figure 1 solid line) with every successive trial increasing the log odds of a correct decision by 0.043 (95% CI 0.019-0.067, $p < 0.001$).

The P_{corr} of a correct choice at the end of the test was higher ($P_{\text{corr}} = 0.905$, 95% CI 0.817-0.954) than at the start of the control ($P_{\text{corr}} = 0.572$, 95% CI: 0.358-0.763), and this difference was significant (log odds ratio = 1.968, $p < 0.001$) (Figure 1, compare error bars between test and control panel).

Also note that P_{corr} is significantly above chance (95% confidence ribbon does not overlap the 50% line) from an early trial number and remains that way until the end of the test. In contrast, the P_{corr} in the control phase is never significantly above chance for the duration of the control (95% confidence ribbon overlaps the 50% line).

That the model suggests naive wasps perform above chance at the onset of the test phase (ie. in trial 1) can be attributed to the fact that the model is unable to fit the rapid learning of

the wasps, rather than to the availability and learning of olfactory and / or temporal cues during the pre-test training phase. This is supported by the proportion of correct decisions in the first few 3-trial wide windows of the test phase (black points) that are clustered around 50%.

Question 2: What information are wasps using to discriminate patterns?

- Do they need optic flow?
- Can they distinguish simple brightness differences?
- are there differences in performance among artificial stimuli?

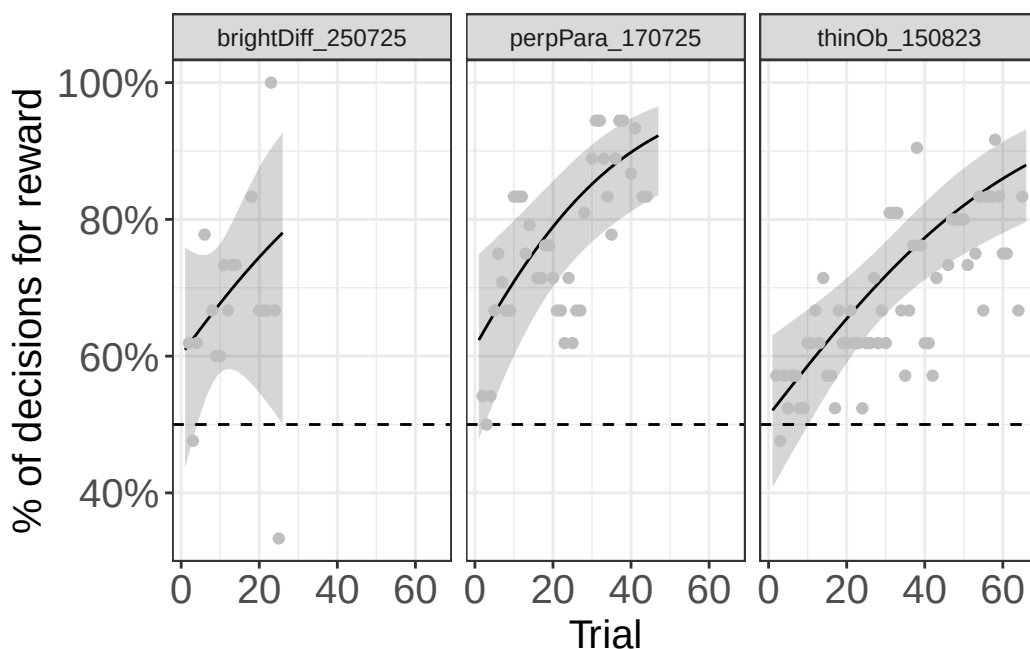


Table 1: Q2 trial-slope summary by artificial experiment

	trialLogOdds	ci95	pFmt	oddsMultiplierPerTrial
brightDiff_250725	0.033	-0.037-0.103	0.356	1.033
perpPara_170725	0.043	0.019-0.067	<0.001	1.044
thinOb_150823	0.029	0.015-0.044	<0.001	1.030

To answer whether wasps require optic flow information to discriminate patterns we tested whether they could distinguish between oblique stripes oriented at 45 degrees to the right or

left of the y-maze arm's long axis and the general direction of wasp travel.

Wasps had no trouble distinguishing between oblique stripes that presumably produce identical rates of optic flow in the dorsal visual field (Figure 2). Similar to the perpendicular-parallel experiment, trial had a significant positive effect on P_{corr} , with every successive trial increasing the log odds of a correct decision by 0.029 (95% CI 0.015-0.044, $p < 0.001$; Table 1). Initially (at trial 1) performance was not significantly different from chance ($P_{\text{corr}} = 0.521$, 95% CI: 0.409-0.631) but rose steeply, reaching 0.880 (95% CI: 0.796-0.932) by the end of the experiment (trial 66), which is significantly higher than expected by chance (50%).

Next, we asked whether wasps could discriminate between an opaque and transparent roof, ie. between stimuli that carried little to no pattern information. While trial had a positive effect on P_{corr} (beta = 0.033, 95% CI -0.037-0.103), unlike for the other two pairs of artificial stimuli this effect was not significant ($p = 0.356$), likely due to the much shorter number of trials this experiment was run for. While statistical support for a positive slope is lacking, there is support for $P_{\text{corr}} > 50\%$ towards the middle and end of the experiment. At the onset of the experiment (trial 1), wasp performance was not significantly different from 50% ($P_{\text{corr}} = 0.610$, 95% CI: 0.436-0.759) but rises with experience and by the end of the experiment (trial 26) P_{corr} reached 0.780 (95% CI: 0.500-0.927). Performance at the end of the experiment is only borderline significantly higher than 50% because of the increased uncertainty in the slope / effect of trials on performance due to the lower number of trials run in this experiment. However, if one uses other metrics or analyses to compare performance to the 50% line, they are significantly different (for example, comparing performance at an intermediate trial or comparing the average performance of the last five trials to the 50% line).

NBNB! Slope or brightdiff exp not significant. fade best fit line?

Comparison among different artificial stimuli:

issues: - day is multicollinear in thin oblique experiment. It makes the slope of thin oblique appear less steep. This drives the difference in slopes detected between perppara and thin oblique. The issue is, is it real? I'm not confident enough that I ran the experiments in exactly the same way so as to make them comparable. If we remove day from the analysis, would we still find a significant difference in the slopes? Could the difference simply be attributable to slower learning due to the side bias which is clearly evident in the thin oblique experiment (possibly caused because of different pre-testing training regimes). Why even is there a side bias when these wasps had just come off the back of a previous experiment with a thick oblique stripe? was the side bias present at that time? - the issue im trying to get at above is that conclusions regarding slope diffs depend on the model you run. - massive confidence intervals around brightness experiment

Question 3: Can wasps use naturalistic patterns for navigation?

- Are performances different from 50%?

- Are performances different from the control?
- Are there differences in performance between naturalistic and artificial stimuli?

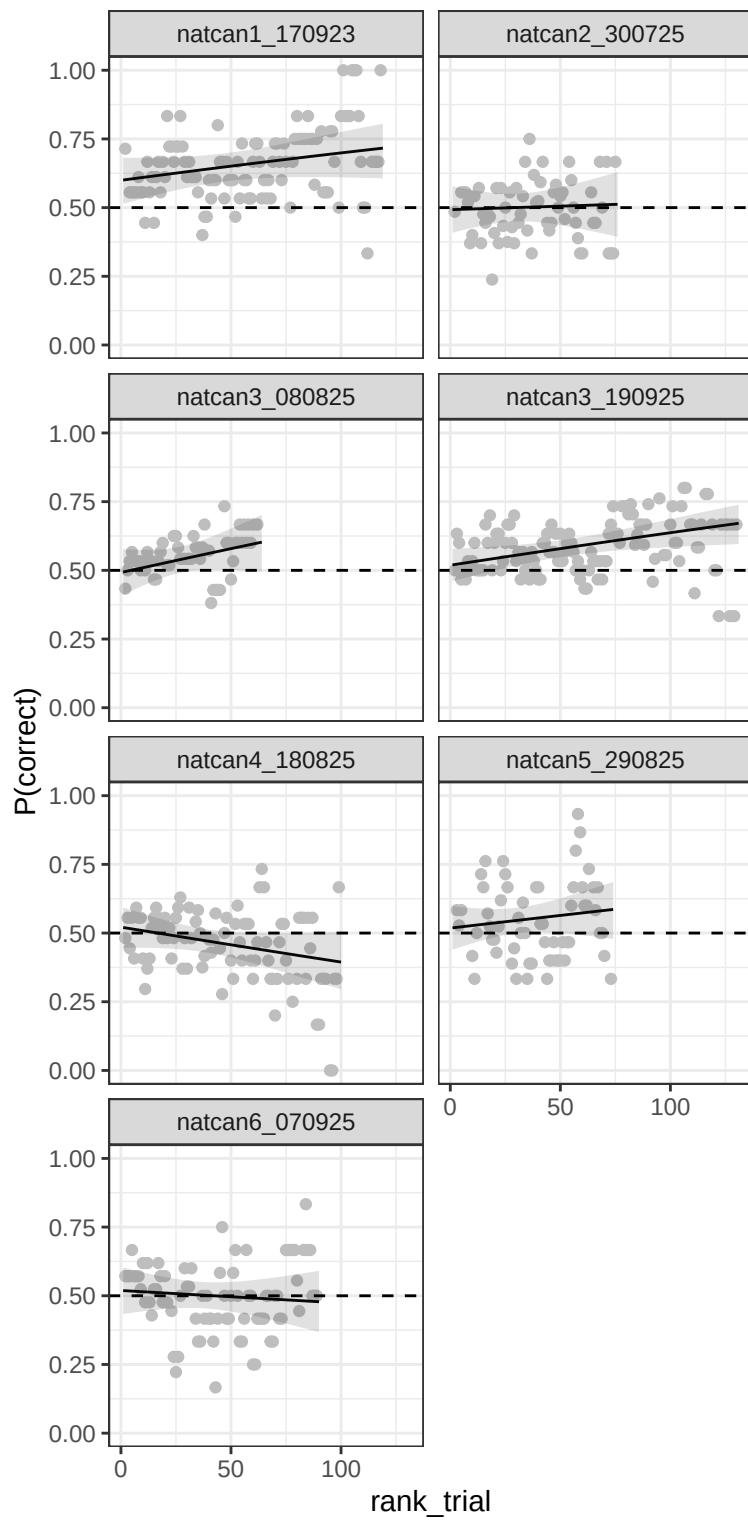


Table 2: Naturalistic test performance at first and last trial

experiment	testStartPcorr95CI	testEndPcorr95CI	testEndTrial	nat_or_art
natcan1_170923	0.600 (0.513-0.682)	0.716 (0.606-0.805)	119	naturalistic
natcan2_300725	0.493 (0.408-0.577)	0.512 (0.394-0.629)	76	naturalistic
natcan3_080825	0.494 (0.411-0.577)	0.603 (0.495-0.701)	64	naturalistic
natcan3_190925	0.520 (0.461-0.578)	0.671 (0.595-0.739)	131	naturalistic
natcan4_180825	0.520 (0.447-0.593)	0.394 (0.296-0.503)	100	naturalistic
natcan5_290825	0.519 (0.439-0.598)	0.586 (0.479-0.686)	74	naturalistic
natcan6_070925	0.519 (0.434-0.603)	0.478 (0.368-0.591)	90	naturalistic

Evidence for whether wasps can discriminate naturalistic canopies at above-chance rates ($>50\%$) is mixed. In four of the seven naturalistic canopy pairs we tested, the predicted P_{corr} by the end of the experiment was significantly above chance (the lower limit of the 95% CI is $> 50\%$, see Table x), suggesting wasps may be able to separate certain pairs of naturalistic canopies to some degree (assuming no other information was available to the wasps).

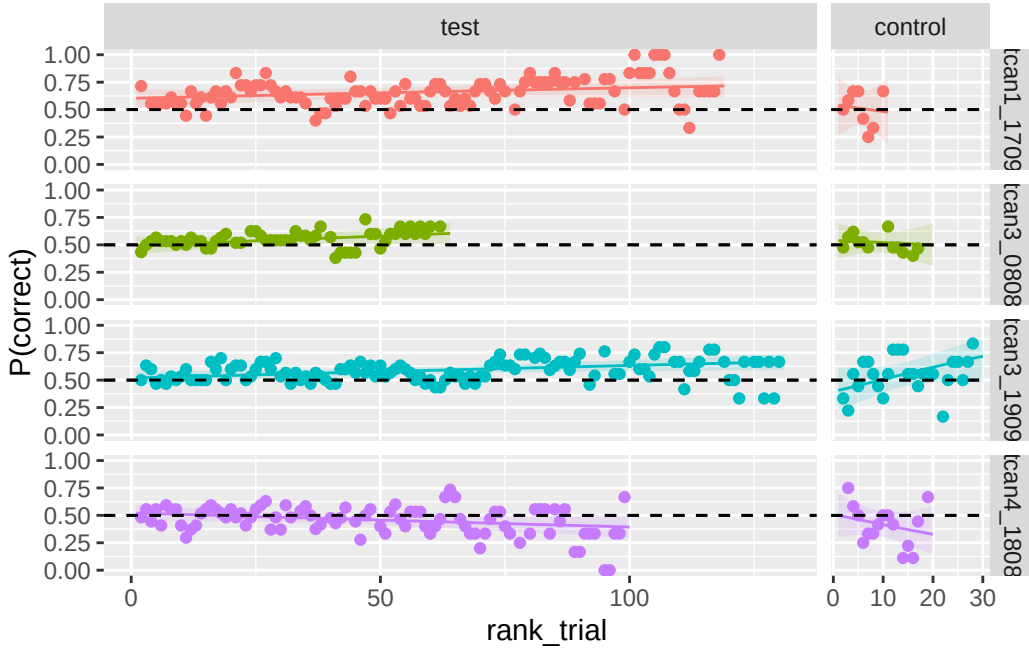


Table 3: Naturalistic test-control performance at test end and control start

experiment	testEndPcorr95CI	controlStartPcorr95CI	testEndTrial
natcan1_170923	0.716 (0.608-0.803)	0.538 (0.263-0.792)	119
natcan3_080825	0.604 (0.500-0.700)	0.540 (0.372-0.699)	64

experiment	testEndPcorr95CI	controlStartPcorr95CI	testEndTrial
natcan3_190925	0.667 (0.591-0.735)	0.406 (0.227-0.615)	131
natcan4_180825	0.394 (0.295-0.502)	0.500 (0.300-0.700)	100

naturalistic test vs control

nat vs art: Discrimination performance for naturalistic canopies tends to be lower than for artificial canopies (make some sort of comparison or just give ranges) and

Produce necessary tables to report from in-line and to include in MS.

Write results here: Evidence that wasps can discriminate naturalistic canopies from the test vs control analysis is also mixed.

Coming soon: 1. comparison of naturalistic stimuli versus artificial

ToDo: - make master tables - add end-trial estimates and CIs to all figures (q2 and q3 outstanding) - x and y figure axis labels - table headers and rownames need to be formatted for pub. - pub-ready experiment names? - eg. perpendicular / parallel and oblique (+45 - 45) - diffuser exp data in supplementary - naturalistic -> artificial transition experiment

Supplementary

	experiment	testStartPcorr95CI	testMidPcorr95CI
1	brightDiff_250725	0.610 (0.436-0.759)	0.705 (0.579-0.807)
2	natcan1_170923	0.600 (0.513-0.682)	0.661 (0.605-0.712)
3	natcan2_300725	0.493 (0.408-0.577)	0.503 (0.448-0.557)
4	natcan3_080825	0.494 (0.411-0.577)	0.550 (0.499-0.599)
5	natcan3_190925	0.520 (0.461-0.578)	0.598 (0.565-0.630)
6	natcan4_180825	0.520 (0.447-0.593)	0.456 (0.409-0.504)
7	natcan5_290825	0.519 (0.439-0.598)	0.553 (0.503-0.602)
8	natcan6_070925	0.519 (0.434-0.603)	0.498 (0.448-0.549)
9	perpPara_170725	0.624 (0.479-0.749)	0.816 (0.731-0.879)
10	thinOb_150823	0.521 (0.409-0.631)	0.741 (0.684-0.791)
	testEndPcorr95CI	middleTrial	testEndTrial
1	0.780 (0.500-0.927)	14	26
2	0.716 (0.606-0.805)	60	119
3	0.512 (0.394-0.629)	39	76
4	0.603 (0.495-0.701)	33	64
5	0.671 (0.595-0.739)	66	131
6	0.394 (0.296-0.503)	51	100
7	0.586 (0.479-0.686)	38	74

8	0.478 (0.368-0.591)	46	90
9	0.923 (0.836-0.965)	24	47
10	0.880 (0.796-0.932)	34	66